YUJEONG JEONG

Environmental AI Modeler

Post-master researcher / National Institute of Green Technology (NIGT)

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RESEARCH INTERESTS

- Knowledge-Guided AI for Environmental Decision Support
- Behavioral Modeling and Reinforcement Learning
- GIS and Satellite-based Monitoring of Land-Carbon Dynamics

EDUCATION

M.S. in Environmental Science and Ecological Engineering — Aug 23, 2024

Korea University, Seoul, Republic of Korea — *(Advisor: Prof. Woo-Kyun Lee)*

Thesis | Spatiotemporal analysis of carbon balance by different administrative levels: Are there carbon neutral local regions in the Republic of Korea? (Master's thesis, Korea University). KERIS Theses & Dissertations. https://dcollection.korea.ac.kr/public_resource/pdf/000000289241_20250511132805.pdf

B.A. & B.E. (Interdisciplinary Program) — Aug 25, 2022

Korea University, Seoul, Republic of Korea

B.A. | French Language and Literature, Department of French Language and Literature, College of Liberal Arts

B.E. | Software Technology & Entrepreneurship Program, College of Informatics

APPOINTMENTS

Oct 2025 – Present — **Post-master Researcher**
Policy Research Division, National Institute of Green Technology (NIGT), Seoul

Sep 2024 – Aug 2025 — **Post-master Researcher**
OJeong Resilience Institute (OJERI), Korea University, Seoul

Mar 2023 – Jul 2024 — **Teaching Assistant**
Dept. of Environmental Science and Ecological Engineering, Korea University, Seoul
- Environmental GIS and Practice (2023 Spring, 2024 Spring); Environmental Remote Sensing and Practice (2023 Fall); Climate Change Impact Modeling (2023 Fall)

Mar 2023 – Dec 2023 — **Research Assistant**
Dept. of Environmental Science and Ecological Engineering, Korea University, Seoul

Sep 2021 – Aug 2022 — **Lab Internship**
Environmental GIS/RS lab, Korea University, Seoul, Republic of Korea

TECHNICAL SKILLS

Modeling	**[AI]** Differentiable / Domain-guided hybrid modeling; CNN and U-Net; Tree-based ML (XGBoost, Random Forest); domain adaptation **[Simulation & Optimization]** Agent-based modeling (ABM); multi-objective optimization (NSGA-II) **[Process-based model]** SWAT; Rothermel-type fire spread models; allometric forest structure models
Geospatial	ArcGIS; satellite and aerial remote sensing; Google Earth Engine
Programming	Python (PyTorch, scikit-learn, GDAL, geopandas, etc.)
Languages	Korean (native); English (Proficient); French (Intermediate; DELF B2)

HONORS, AWARDS & FELLOWSHIPS

2025 — **Best Poster Award**
Summer Conference of Korean Society of Forest Science (KSFS)

2023 — **Best Paper Award**
「BK21PLUS Eco-Leader Keystone Symposium」

2022-2023 — **Graduate Research Scholarship**
「Brain Korea 21 Plus」(BK21+)

2019 — **Grand Prize**
「2019 Forestry ICT Contest by Korea Forest Service」

PUBLICATIONS & PATENTS

Peer-reviewed Journal Articles (SCI & KCI)

2026

Jeong, Y., Jo, H. W., Lee, W. K. (n.d.) Watershed-based FWB (Food-Water-Biodiversity) Nexus Assessment using SWAT-ABM: A case study in the Yeongsan River Basin of the Republic of Korea. Manuscript in preparation.

Jeong, Y., Jo, H. W., Lee, W. K. (n.d.) A Regional Carbon Balance and Typology Map to Support Local Carbon Neutrality Planning in the Republic of Korea. Under review in Journal of Korean Society of Forest Science.

Jeong, Y., Lee, J., Park, C. (n.d.). Development of a Quantitative Framework for Assessing Climate Technology Cooperation Potential between Korea and Developing Countries. Under review in Journal of Climate Change Research (J-2026-077-20260722).

Jeong, Y., Jo, H. W., Roh, M., Kim, S., Lee, W. K. (n.d.). A Hybrid Allometric-Machine Learning Model for Crown Base Height Prediction in the Republic of Korea. Under review in Ecological Informatics (ECOINF-D-25-04526R1).

2024

Ko, Y. J., Song, C. H., **Jeong, Y.**, Hong, M., Kim, J., Lee, W. K. (2024). Development of village-level forest carbon sink map for spatial carbon sink management of the local government. Journal of Climate Change Research, 15(6), 989-1000. https://doi.org/10.15531/KSCCR.2024.15.6.989

Jeong, Y., Song, C. H., Jo, H. W., Ko, Y. J., Lee, W. K. (2024). Development and applicability assessment of local-scale carbon emission models to support carbon neutrality plans of local governments. Journal of Climate Change Research, 15(5-1), 691-721. https://doi.org/10.15531/KSCCR.2024.15.5.69

Kim, J., Jo, H. W., Kim, W. J., **Jeong, Y.**, Park, E. B., Lee, S. J., Kim, M. L., Lee, W. K. (2024). Application of the domain adaptation method using a phenological classification framework for the land-cover classification of North Korea. Ecological Informatics, 81, 102576. https://doi.org/10.1016/j.ecoinf.2024.102576

Conference Presentations

2025 **Jeong, Y.** (2025, August). Hybrid Allometric–Machine Learning Model for Predicting and Mapping Crown Base Height in Forests of the Republic of Korea [Conference Presentation (Poster)]. Korean Society of Forest Science (KSFS), Republic of Korea.

Jeong, Y. (2025, May). Watershed-based FWB (Food-Water-Biodiversity) Nexus Assessment using SWAT-ABM: A Case Study in the Yeongsan River Basin of the Republic of Korea [Conference Presentation (Oral)]. European Geosciences Union (EGU), Vienna, Austria.

2024 **Jeong, Y.** (2024, April). Are There Carbon-Neutral Cities in South Korea?: Using Residual Modeling on Different Spatial Scales [Conference Presentation (Oral)]. European Geosciences Union (EGU), Vienna, Austria.

2023 **Jeong, Y.** (2023, December). Are There Carbon-Neutral Cities in South Korea?: Spatial Analysis of Carbon Balance on Different Scales [Conference Presentation (Oral)]. American Geophysical Union (AGU), San Francisco, USA.

Jeong, Y. (2023, April). Assessing the Sustainability in Water Use under Different Agricultural Management Planning in Yeongsan-River Basin, South Korea [Conference Presentation (Oral)]. European Geosciences Union (EGU), Vienna, Austria.

2022 **Jeong, Y.** (2022, December). Analysis on Carbon Sources and Sinks for the Achievement of Carbon Neutrality in South Korea: Case study on Ulsan [Conference Presentation (Poster)]. Korean Society of Climate Change Research Annual Conference, Jeju, Republic of Korea.

Patents

2025 Lee, W. K., Cha, S. E., Kim, S., Jo, H. W., **Jeong, Y.**, Roh, M. (2025). System and Method for Predicting Wildfire Spread Using Artificial Intelligence and Process-Based Simulation. Korean Patent Application No. 10-2025-0189148, filed Dec. 3, 2025. Patent pending.

Software & Data

2026 AI-Allometric Hybrid Model for Crown Base Height(CBH) Mapping — trained model, maps and training data. Zenodo, DOI: 10.5281/zenodo.21736584.

Carbon Balance prediction model — local-scale carbon balance model, source code, maps and training data. github.com/mongbleu/carbonBalance.

Climate Technology International Cooperation Strategy Map — web platform, public deployment. ctis.re.kr/map.

Policy Briefs & Reports

2025 UN ESCAP, et al. (2025) Scaling the Carbon Neutral Village model for rural communities' climate action. Retrieved from: https://hdl.handle.net/20.500.12870/8214.

2024 Climate Technology Centre and Network (CTCN). (2024). Output 3.1-3.6 synthesis report. https://www.ctc-n.org/system/files/dossier/3b/Output%203.1-3.6%20Synthesis%20Report.pdf

2025 Lee, W. K., et al. (2025). Development of territorial spatial planning and management technology to reduce GHG emissions (Report No. TRKO202500010558). Ministry of Land, Infrastructure and Transport. https://scienceon.kisti.re.kr

RESEARCH PROJECTS

Crown Base Height and Crown Fire Prediction Module | Korea Forestry Promotion Institute
Assistant Researcher | Apr 2024 – Dec 2025

- **Key Outcomes**
  - [Journal] 1 article at Ecological Informatics (under major revision, first author).
  - [Patent] KR 10-2025-0189148, filed Dec 2025.
  - [Data & Code] Open dataset and trained model (Zenodo DOI: 10.5281/zenodo.21736584)
  - [Conference] KSFS 2025, poster (Best Poster Award).
- **Core Contributions**
  - Developed a crown base height prediction and crown fire simulation module for 36 tree species.
  - Converted a physics-based fire-spread model into a differentiable CNN layer.
  - Produced training data for the forest fire propagation model by calculating Normalized Burn Ratio (NBR) from Sentinel imagery and analyzing observed GIS fireline data.

Local-Scale Carbon Balance Model and Maps of the Republic of Korea | Environmental GIS/RS Lab, Korea University
Assistant Researcher | Sep 2022 – Aug 2024

- **Key Outcomes**
  - [Journal] 2 articles at Journal of Climate Change Research (2024, first author on one).
  - [Journal] 1 article at Journal of Korean Society of Forest Science (under review, first author).
  - [Code] Open model repository (github.com/mongbleu/carbonBalance)
  - [Thesis] M.S. thesis, Korea University (2024).
  - [Conference] EGU 2024, AGU 2023, KSCCR 2022.
- **Core Contributions**
  - Built an AI-driven spatial model predicting greenhouse gas emissions at the finest administrative level.
  - Developed the local-scale carbon balance maps and its typology map.

SWAT-ABM(Agent-Based Model) Coupled Model for the Food-Water-Biodiversity Nexus (fairSTREAM project) | National Research Foundation of Korea · IIASA Strategic Initiatives
Assistant Researcher | Sep 2021 – Sep 2024

- **Key Outcomes**
  - [Journal] 1 manuscript in preparation (first author).
  - [Conference] EGU 2025, EGU 2023.
- **Core Contributions**
  - Coupled a process-based hydrological model with an NSGA-II behavioral model, simulating one million farming households under climate and management scenarios.

Climate Technology Cooperation Strategy Map | Ministry of Science and ICT · National Institute of Green Technology
Researcher | Jan 2026 – Dec 2026

- **Key Outcomes**
  - [Platform] Deployed public version for R&D cooperation with developed countries (website: ctis.re.kr/map)
  - [Platform] Deployed internal version for developing countries
  - [Journal] 1 article at Journal of Climate Change Research (under review, first author).
- **Core Contributions**
  - Designed the analytical framework for the strategy map and its platform, covering cooperation with both developing and developed countries.

Land Cover Mapping for the National GHG Inventory of Korean Ecosystem Types | Ministry of Environment

Assistant Researcher | Oct 2024 – Present

- **Key Outcomes**
  - [Journal] 1 article at Ecological Informatics (2024, co-author).
- **Core Contributions**
  - Developing a nationwide land cover map of the Republic of Korea using a U-Net model and satellite imagery.

Carbon Neutral Village Model for Rural Nepal | UN ESCAP

Assistant Researcher | Sep 2024 – Feb 2025

- **Key Outcomes**
  - [Report] UN ESCAP (2025), Policy brief and Report.
- **Core Contributions**
  - Designed the carbon neutral village concept and action plan; estimated mitigation and adaptation outcomes from field data collected in rural Nepal.

Samoa Forest Carbon Detection and Quantification | National Institute of Green Technology

Assistant Researcher | Apr 2023 – Jan 2024

- **Key Outcomes**
  - [Report] CTCN (2024), Synthesis Report.
- **Core Contributions**
  - Co-developed ML- and DL-driven land cover classification and a satellite imagery processing pipeline.

National GHG Inventory of the LULUCF Settlement Sector | Korea Agency for Infrastructure Technology Advancement · Ministry of Land, Infrastructure and Transport

Assistant Researcher | Apr 2020 – Dec 2024

- **Key Outcomes**
  - Lee, W. K. et al. (2025). *Development of territorial spatial planning and management technology to reduce GHG emissions* (Report No. TRKO202500010558). Ministry of Land, Infrastructure and Transport. kisti.re.kr
- **Core Contributions**
  - Analyzed aerial imagery to develop spatially-explicit activity data of the national greenhouse gas inventory for the Settlement sector under LULUCF
  - Co-authored Chapter 2.2 of the report focusing on land-use change detection and carbon accounting methodologies.

EXTRA-CURRICULAR ACTIVITIES

2025 **Speaker at the Seminar**
「The 1st Carbon Neutral Village Network Seminar: Carbon Neutral Village in Mid-Latitude Region」, OJeong Resilience Institute, Korea University, Seoul, Republic of Korea

2024 **Consultation**
Research project 「Establishing the Foundation for Technological Advancement and Promoting the Use of Carbon Maps Based on CO_2 and CH_4 Measurements」 of Climate Change Center, Republic of Korea

2023 **Member**
Media team & COP23 TF, GEYK (Green Environment Youth Korea), Seoul, Republic of Korea

2022 **AI Bootcamp Course**
Code States, Seoul, Republic of Korea

Chief Editor
Climate change magazine「WAY:D」, Seoul, Republic of Korea

2020 **Internship program KOICA YP** (Young Professional)
Korea Green Fund, Seoul, Republic of Korea

Student voluntary service project designer
KUSSO (Korea University Service Organization), Korea University, Seoul, Republic of Korea
- Hosted IECU (International Environmental Conference Uzbekistan) & Online workshops with Uzbekistan undergraduate students for Aral Sea Recovery
- Environment Education Program for the children of multicultural family

Others

- Volunteer French interpreter, World Taekwondo Peace Corps Foundation (TPC), Seoul, Republic of Korea, (Algeria, Aug 2018 & Gabon, Jan 2020)
- The First Place, Women's -57kg, 2020 National University Taekwondo Club Competition